

Charles Wang

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EDUCATION

Purdue University, M.S., Agricultural & Biological Engineering

Jan 2026 – May 2027 (expected)

Purdue University, B.S., Biological Engineering · GPA 3.63

Dec 2025

EXPERIENCE

Hardware Engineer, LeafSpec LLC / Purdue ABE Plant Sensor Lab

May 2024 – Present

Purdue ag-tech startup (Prof. Jian Jin) commercializing handheld leaf imagers for farmers and researchers

- Engineer handheld crop-imaging instruments from concept to field deployment: designing enclosures and mechanisms in SolidWorks, prototyping via 3D printing, and integrating cameras, LED illumination, motors, and microcontrollers into devices used daily by plant researchers.
- Build the software that runs each instrument alongside the hardware, from web control consoles to Android apps, including LLM-based analysis that detects plant stress and disease signals from leaf images.
- Ship hardware with real-world adoption: my LeafDoc capture hardware supports a **Purdue-DARPA project** using plants as chemical sentinels, and lab imagers are used by partners including **Corteva**.

Summer Research Fellow, Institute for Digital Forestry, Purdue University

May – Jul 2024

Paid summer research program; placed in the ABE Plant Sensor Lab

- Rebuilt the lab's undocumented LeafSpec-Multi multispectral imager from its SolidWorks assembly alone: tore down an undocumented legacy unit to learn how it worked, 3D-printed every part, and rebuilt the complete device.
- Hand-soldered a **120+ LED illumination board** spanning 6 wavelength bands (visible + IR) for multispectral imaging, with MOSFET drivers per band on Raspberry Pi GPIO and a 12 V → 5 V power converter.
- Designed a new top enclosure in SolidWorks that moved the Pi, driver board, and converter from an external tethered box into the device body, making it **~30% smaller** and truly handheld, and added GPS/GIS geo-referenced capture via app integration.

SELECTED PROJECTS: PLANT SENSOR LAB

Leaf Cross-Section Microscope Imager · M.S. Project

Jan 2026 – Present

- Prototyping a modular attachment for the lab's handheld 400× leaf microscope that images full corn-leaf cross sections for nitrogen sensing, adding an internal dimension of analysis beyond surface-level hyperspectral imaging.
- Designed a motor-driven clamp and stage that steps the leaf laterally through the optical path for frame-by-frame capture at 200×; replaced the gear-driven focus stage with an electrically tunable liquid lens for fast, vibration-free focusing.
- Built a web-based control console (lighting, motion, live view, capture) and a desktop app that stitches frames into one continuous cross-section image with alignment and color correction.

LeafDoc, Smartphone Leaf-Health System

Jun 2025 – Present

- Lead a team of 6 undergraduate engineers building a smartphone-based system for plant leaf health analysis: a compact leaf-isolating lightbox with rollers that flatten corn leaves for evenly lit, repeatable captures; validated on nearly 1,000 leaves, cutting capture time from ~10 s to **under 3 s per leaf**.
- Built the Android capture app and cloud pipeline: guided capture, labeling and sorting, multi-frame stitching into full-length panoramic leaf images, and a prototype LLM diagnosis feature returning a plant health score.

FluoroScan, Chlorophyll Fluorescence Imager

Sep 2024 – May 2025

- Engineered an imager that measures how efficiently plants photosynthesize for **under 1% of the cost** of commercial instruments, using twelve 450 nm excitation LEDs delivering 4,000 $\mu\text{mol m}^{-2} \text{s}^{-1}$ of saturating light, capturing 680 nm chlorophyll emission.

SKILLS

Embedded & Electronics	Raspberry Pi, GPIO, MOSFET LED drivers, soldering, motor control, power conversion
Imaging & Optics	Hyperspectral, multispectral, and fluorescence imaging; microscopy; liquid-lens focusing
CAD & Fabrication	SolidWorks parts and assemblies; rapid prototyping; hardware debug and test
Software	Python, C, MATLAB; Android; image stitching; LLM APIs and AI-assisted development
Languages	English, Chinese (fluent)

HONORS & LEADERSHIP

- Top 10, Indiana Soybean Innovation Competition (2024): co-created Soysheets, a 67% soy detergent and dryer sheet.
- Executive Board, Association of Hip Hop Practitioners & Scholars (2021–Present): led a fundraiser that raised \$80,000.